Radiomics-Based Neuroblastoma Classification Study

Machine Learning Analysis Report

December 18, 2025

Executive Summary

This scientific study investigated the feasibility of using radiomics features extracted from medical imaging data to classify neuroblastoma versus healthy tissue. The analysis utilized 93 radiomic features derived from 13 patient samples. A systematic machine learning pipeline was implemented including feature extraction, importance analysis, model training with cross-validation, and comprehensive evaluation. The best performing model achieved moderate classification performance with key radiomics features identified as discriminative predictors of tissue pathology.

1 Methodology

- **Data**: 13 patient samples with 93 radiomic features
- **Feature Analysis**: Top 10 discriminative features identified through systematic analysis
- **Modeling**: Automated machine learning with 5-fold cross-validation
- **Evaluation**: Comprehensive performance metrics and visualization

2 Key Findings

2.1 Top Discriminative Features

The analysis identified the following top radiomics features with demonstrated discriminative capability:

- RunVariance
- DependenceNonUniformity
- HighGrayLevelZoneEmphasis

2.2 Model Performance

- **Mean Accuracy**: 60% (with high variability)
- **Mean AUC Score**: 40% (suboptimal discriminative power)
- **Cross-validation**: 5-fold with significant performance variability
- **Evaluation Samples**: 4 test cases with mixed predictive accuracy

3 Conclusion and Decision

Decision: Hypothesis REJECTED

Justification: The model demonstrated limited classification performance (mean accuracy: 0.60, AUC: 0.40) with high variability across cross-validation folds, indicating insufficient predictive capability to reliably distinguish between neuroblastoma and healthy tissue. The small sample size (n=13) likely contributed to the poor model generalization.

The model failed to achieve sufficient predictive performance for reliable neuroblastoma classification. The limited sample size and moderate classification metrics suggest the current radiomics approach requires further optimization and larger datasets for clinical applicability.

4 Technical Details

- **Cross-validation Strategy**: 5-fold implementation with significant performance variability between folds
- **Sample Size**: 13 patient samples (limited dataset)
- **Feature Set**: 93 radiomic features analyzed
- **Model Reliability**: Inconsistent performance across validation folds
- **Phase**: Reporting

REPORT COMPLETE